# Exit Ticket · reflect and retrieve

## Exit Ticket

A touchscreen can only be an output device.

## Use key vocabulary from memory

1. Explain input in your own words.

2. Explain output in your own words.

3. Explain sensor in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 103 · Input and Output: Selecting Devices for Tasks

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What does an input device do?

### Q2

What does an output device do?

### Q3

What is an actuator?

### Q4

A 2×3 display grid: how many cells?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Understand — An input device captures data or a user action; an output device communicates information or changes physical conditions.

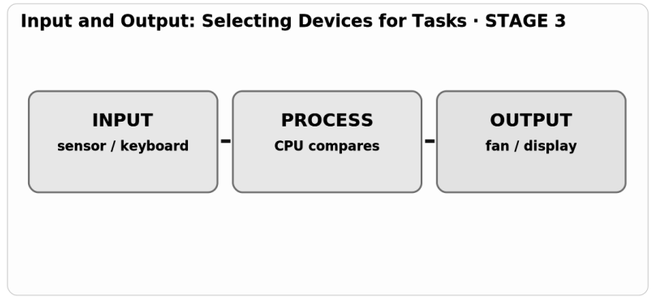
## Core idea 2

Apply — A temperature sensor measures physical conditions; an actuator controls a physical action; a display shows values to the user.

## Core idea 3

Evaluate — Choose devices by required accuracy, accessibility, environment and response time. A touchscreen can perform both input and output functions.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Design accessible touchscreen ticket kiosk: give two input and two output features and justify. [8]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

For a room-temperature controller, choose appropriate input and output devices and explain roles. [6]

## Question 2

Compare barcode reader and RFID for counting library books, linking benefits to context. [6]





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

input — Data entering computer system.

output — Information/effect produced.

sensor — Measures a physical quantity.

actuator — Creates a physical action.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

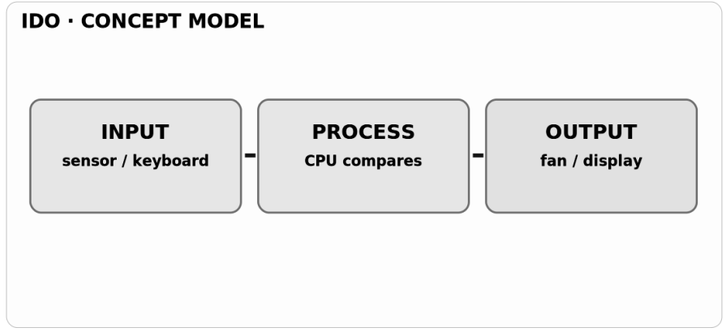
## Problem

Recommend input and output for automated greenhouse temperature control.

## Follow the worked method

1. Explain the choice — Input thermistor/temperature sensor; output fan/heater actuator, possibly a display.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Explain the choice
Input barcode reader or RFID reader; output display and optional printer.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.